TAMIL NADU STATE BOARD - STANDARD 10 SCIENCE

CHAPTER 1: PHYSICS

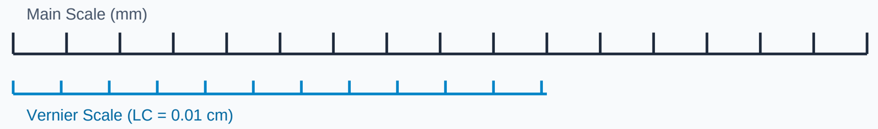
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Science
Chapter Name:	Physics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division



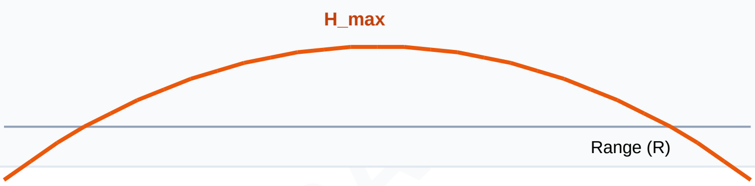
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Inertia	The inherent property of a body to resist any change in its state of rest or state of uniform motion, unless acted upon by an external force.
Impulse	A large force acting for a very short duration, measured as the product of the force and the time interval, equal to change in momentum.
Power of Lens	The degree of convergence or divergence of light rays achieved by a lens, calculated as the reciprocal of its focal length in meters.
Ohm Law	At constant temperature, the steady current flowing through a conductor is directly proportional to the potential difference across its ends.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

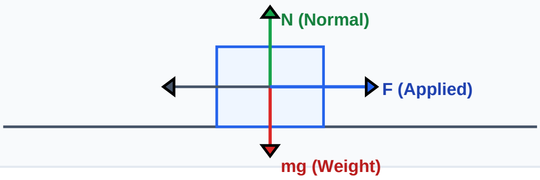
Kinematic Projectile Trajectory



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Force: F = m * a | Linear Momentum: p = m * v
- Lens Formula: 1/f = 1/v - 1/u | Power: P = 1 / f (in diopters)
- Ohm's Equation: V = I * R | Joule Heat: H = I^2 * R * t
- Resistance (Series): Rs = R1 + R2 + R3
- Resistance (Parallel): 1/Rp = 1/R1 + 1/R2 + 1/R3

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Calculate the equivalent resistance of three resistors of 2 ohms, 3 ohms, and 6 ohms connected in parallel.

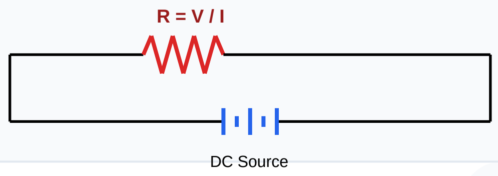
Step-by-step Solution:

Apply parallel resistance formula: 1/Rp = 1/2 + 1/3 + 1/6.

Find common denominator (6): 1/Rp = (3 + 2 + 1) / 6 = 6 / 6 = 1.

Therefore, equivalent resistance Rp = 1 ohm.

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

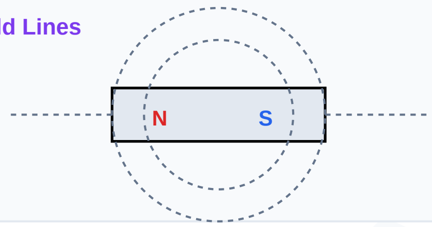
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: An object is placed at a distance of 20cm in front of a convex lens of focal length 10cm. Find the position and nature of the image.

Analytical Solution Strategy:

Here, object distance u = -20 cm, focal length f = +10 cm.
Lens formula: 1/v - 1/u = 1/f => 1/v - 1/(-20) = 1/10 => 1/v + 1/20 = 1/10.
1/v = 1/10 - 1/20 = 1/20 => v = +20 cm.
The image is formed at 20cm on the other side of the lens. It is real and inverted, and same size since u = 2f.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Newton's Second Law of Motion.

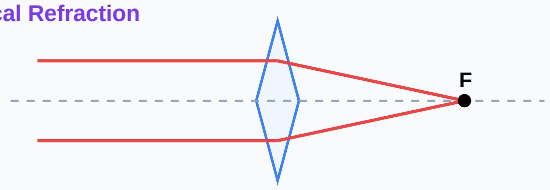
Q2 (2-Mark): Define refractive index of a medium.

Q3 (2-Mark): Write the equations representing Boyle's Law and Charles's Law.

Q4 (5-Mark): A current of 2A flows through a resistor of 5 ohms for 10 seconds. Find the heat generated.

Q5 (5-Mark): Explain the causes of myopia (short-sightedness) and how it is corrected.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Home Wiring Systems

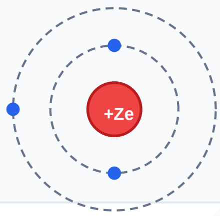
Household appliances are wired in parallel circuits to ensure each device receives standard 220V voltage input.

Application Case: Optical Diagnostics

Ophthalmic centers use lens power formulas to manufacture prescription eyeglasses correcting myopia.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Sinusoidal AC Voltage Waveform